



Transient mixed convection heat transfer for opposing flow from two discrete flush-mounted heaters in a rectangular channel of finite length: Effect of buoyancy and inclination angle



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ABSTRACT

An experimental investigation in a vertical rectangular channel using water as the working fluid is carried out to study the transient laminar opposing mixed convection heat transfer from two flush-mounted, symmetric and discrete heat sources subjected to a constant wall heat flux boundary condition while the other bounding walls are insulated and adiabatic. The experiments are done under different values of buoyancy strength or modified Richardson number $Ri^* = Gr^*/Re^2$, Reynolds number of $300 \leq Re \leq 900$ and channel inclination of $0^\circ \leq \gamma \leq 90^\circ$. From experimental measurements, surface temperature distributions and averaged Nusselt number for each heat source are obtained. In general, for a fixed value of the buoyancy parameter, the averaged Nusselt number increases for increasing values of the Reynolds number. In the vertical channel configuration, it is observed that for fixed values of Re and high Ri^* number, because buoyancy acts directly against convective flow, higher heat transfer rates are achieved. As the duct approaches the horizontal configuration, buoyancy strength is reduced and the averaged Nusselt number decreases for decreasing values of the inclination angle with marked variations. Here, the effect on the heat transfer rates is more pronounced at $\gamma = 60^\circ$ for low Ri^* . For the horizontal configuration, because buoyancy only acts indirectly, higher threshold values of Ri^* are required to induce instability. The results show that for relatively large values of buoyancy strength, the surface temperature presents strong spanwise and axial variations, and for all of the inclination angles considered in this study, the values of the surface temperatures achieve higher values at the middle spanwise positions of both heaters than those registered at other spanwise locations. This indicates that because of the secondary three-dimensional flow, heat transfer augmentation takes place close to the channel corners while the higher surface temperatures and hence, lower heat transfer rates are achieved at the centerline of the discrete heat sources.

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1. Introduction

Mixed convection heat transfer in the presence of finite-size heat sources has become a subject of increased interest because of advances in the electronics industry and heat exchanger technology. The outcome of advances in electronic systems

miniaturization is that increased heat flux densities are obtained, which has generated and increased need for dependable and efficient cooling technologies. Because the reliability and durability of electronic devices depends on their capability to dissipate excessive heating, numerous theoretical and experimental investigations aimed to increase power dissipation of these systems are available in the literature. However, although mixed convection flow in vertical ducts is inherently three-dimensional (3D), the majority of the studies available are analyzed in two dimensions [1–9], and relatively few investigations of flow reversal of 3D flow and heat transfer have been reported [10–15]. Dogan et al. [16] investigated experimentally the mixed convection heat transfer from discrete

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Nomenclature

A_c	cross-sectional area of the channel
A_{heater}	surface area of each heater exposed to the water flow
D_H	hydraulic diameter (characteristic length)
F	nondimensional function defined in Eq. (A.13)
f	frequency (Hertz)
G	nondimensional function defined in Eq. (A.13)
g	gravity acceleration
Gr^*	modified Grashof number, $Gr^* = g\beta\dot{q}_{conv}D_H^4/k\nu^2$
h	heat transfer coefficient
H	channel width
I	measured current
k	fluid thermal conductivity
l_2	length of the discrete heat sources
L_i	nondimensional length of the discrete heat sources, $L_i = l_i/D_H$
Nu	local Nusselt number based on the hydraulic diameter
$\overline{Nu}$	space-averaged Nusselt number based on the hydraulic diameter
$\widetilde{Nu}$	time-and-space averaged Nusselt number
P	channel perimeter
Pr	Prandtl number, $Pr = \nu/\alpha$
$\dot{Q}$	net convective heat flux transferred to the fluid
$\dot{q}$	net convective heat flux per unit surface transferred to the fluid
$\dot{q}_{cond}$	calculated conduction losses to the ambient
$\dot{q}_{el}$	measured input power per unit surface supplied to each heater
$\dot{q}_{rad}$	calculated radiation losses to the ambient
Re	Reynolds number based on the hydraulic diameter, $Re = u_0D_H/\nu$

Ri^*	modified Richardson number, $Ri = Gr^*/Re^2$
St	Strouhal number based on the hydraulic diameter, $St = fD_H/u_0$
t	time
T_{amb}	ambient temperature
T_0	fluid temperature at the channel inlet
T_w	local surface temperature
$\bar{T}_w$	mean surface temperature of each heater
u_0	fluid velocity at the channel inlet
V	measured voltage
x, y, z	rectangular Cartesian coordinates
X	nondimensional axial coordinate, $X = x/D_H$
Y	nondimensional transverse coordinate, $Y = y/D_H$
Z	nondimensional spanwise coordinate, $Z = z/D_H$

Greek symbols

α	thermal diffusivity
β	thermal volumetric expansion coefficient
ΔT	temperature difference, $\Delta T = T_{wj} - T_0$
ε	surface emissivity of aluminum
γ	inclination angle with respect to the horizontal
λ	parameter defined in Eq. (A.13)
ν	kinematic viscosity
ρ	fluid density
σ	Stefan–Boltzmann constant, $5.670373(21)10^{-8} \text{ W/m}^2 \text{ K}^4$
τ	nondimensional time

Subscripts

$j = R, L$ indicates left and right heated slabs, respectively

heat sources inside a horizontal rectangular channel and presented variations of the local Nusselt number and heater surface temperature. Bhowmik et al. [17] performed experiments to study the heat transfer characteristics on an array of four in-line, flush-mounted simulated chips in a vertical up-flow rectangular channel during steady-state operation and developed a single correlation to describe the convection modes. Gau et al. [18] studied experimentally the reversed flow structure and heat transfer in a finite parallel-plate channel with one wall heated uniformly and the opposite wall insulated. They pointed out that when Gr/Re^2 is greater than a threshold value, the flow is highly unstable. Dutta et al. [19] performed experimental measurements of buoyancy-assisted mixed convection in a vertical square channel with an asymmetric heating condition. Their results show that opposed flows have higher heat transfer coefficients and observed large 3D scale flow structures in the buoyancy affected flow. Zhang and Dutta [20] used the same experimental setup used in Ref. [19], but applied different heating conditions for a range of Reynolds numbers from 200 to 11,200 and a range of the buoyancy parameter, Gr/Re^2 from 0.02 to 200. Cheng et al. [21] performed numerical predictions of developing flow with buoyancy-assisted flow separation in a vertical rectangular duct using a 3D parabolic, boundary-layer model and the FLARE approximation. Their results show that the strength and extent of the reversed flow are dependent mainly on the ratio Gr/Re and the aspect ratio of the cross section, and that the threshold value of Gr/Re for the occurrence of flow reversal becomes higher when the value of the aspect ratio is increased. The same authors [22] assessed the validity and efficiency of their

parabolic model for analyzing 3D flow separation by comparing it against a full elliptic Navier–Stokes model under various heating conditions, Prandtl numbers, aspect ratio of the cross section of the duct, and buoyancy parameter. Cheng et al. [23] studied the 3D mixed convection in a vertical enclosure with buoyancy assisted through-flow with various asymmetric heating conditions and predicted the criteria for the flow reversal to occur. Sudhakar et al. [24] reported results of a numerical investigation of the problem of finding the optimum configuration for five discrete heat sources mounted on a wall of a 3D vertical duct under mixed convection heat transfer using artificial neural networks. Martínez-Suástegui and Treviño [25] investigated experimentally the transient laminar mixed convection in an asymmetrically and differentially heated vertical channel of finite length subjected to an opposing buoyancy. Their results show that for values of the Richardson number larger than a critical one, the flow patterns are nonsymmetric, periodic, and exhibit increasing complexity and frequency for increasing buoyancy.

Although most studies found in the literature have focused on the horizontal or vertical configuration for different channel geometries, boundary and operating conditions [26–28], in many practical situations the duct is inclined with respect to the horizontal. Because the flow and heat transfer are relatively sensitive to duct orientation, a great deal of research efforts have been devoted to this topic. Guimarães and Menon [29] numerically studied mixed convection in an inclined rectangular channel with three discrete heat sources placed on the bottom surface. Their results show that in general, cases which show the lowest

temperature distribution on the modules are those where the inclination angles are 45° and 60° . Choi and Ortega [30] numerically investigated the effects of inclination angle during mixed convection for a parallel plane channel with a discrete heat source. Their results indicate that the overall Nusselt number strongly depends on the inclination angle when the latter is larger than 45° , while the changes in the maximum surface temperature and Nusselt number are negligible when the inclination angle is from 0° to 45° . Ichimiya and Matsushima [31] performed numerical simulations to study the 3D effect of the inclination angle on the mixing flow in a square channel with uniform temperature walls and inlet temperature. The authors reported that high heat transfer and low pressure loss area was obtained for inclination angles of -15° to -60° . Chong et al. [32] studied experimentally the flow and heat transfer in an inclined and horizontal rectangular duct with a heated plate longitudinally mounted in the middle of the cross section and subjected to a uniform heat flux for $Re = 334$ – 1991 , $Gr = 5.26 \times 10^3$ – 5.78×10^6 and inclination angles of -60° to 60° . They pointed out that the heat transfer coefficients and pressure drops were independent of inclination angles when the Reynolds number increased to about 1800. Morcos et al. [33] and Maughan and Incropera [34] studied experimentally the effect of surface heat flux and channel orientation and provided experimental data for the local and average Nusselt numbers. Huang and Lin [35] performed numerical simulations of 3D mixed convection air flow in a bottom heated inclined rectangular duct. They delineated the effects of duct inclination on the flow transition and the associated temporal and spatial flow structures, and showed that increasing the inclination angle does not always stabilized the flow. Ozsunar et al. [36,37] numerically and experimentally investigated mixed convection heat transfer in rectangular channels under various operating conditions and assessed the effects of channel inclination, surface heat flux and Reynolds number on the local Nusselt number distributions. They pointed out that the onset of instability was found to move upstream for increasing Grashof number and delayed for increasing Reynolds number and increasing inclination angle. Chong et al. [38] carried out experiments to study the mixed convection heat transfer of thermal entrance region in an inclined rectangular duct for laminar and transition flow and determined the effects of inclination angles on the heat transfer coefficients and friction factors. Lin and Lin [39,40] performed experimental measurements for the mixed convective air flow through a slightly inclined rectangular duct for opposing buoyancy and pointed out that at a higher buoyancy-inertia ratio, the reversed flow moves upstream and becomes either time periodic or quasi-periodic.

The above review of literature suggests that little attention has been paid to evaluate the buoyancy and channel orientation effects on the space-averaged Nusselt number distributions from two discrete flush-mounted heaters placed symmetrically in a rectangular channel. Also, transient 3D flow behavior has been given very little attention in both numerical and experimental studies, and temperature fluctuations due to flow oscillation have been practically overlooked. The aim of the present study is to conduct a detailed experimental study to investigate the 3D effects of opposing buoyancy and channel orientation on the local and spaced-averaged surface temperature distributions of each module. The results include the effects of the Reynolds and Richardson numbers on the evolution of the final flow and thermal response and overall Nusselt numbers. In this paper, the experimental set-up used to obtain the experimental data is described. The calculation procedure established to determine the local and averaged mixed convection heat transfer coefficients and the dimensionless data analysis is explained. Finally, the

results of the experimental data are presented and discussed in detail.

2. Materials and methods

In this section, information about the experimental apparatus, devices used and procedures followed for the processing of experimental data are provided.

2.1. Experimental arrangement

A schematic of the experimental set-up used for the present study is shown in Fig. 1. The nozzle section and test section are fixed on a metal framework for rigidity purposes, and the whole channel can be kept at any prescribed angle from $0^\circ \leq \gamma \leq 90^\circ$ by the chassis. A 3D Cartesian coordinate system is selected such that x , y and z -axes are set along the streamwise, wall-normal, and spanwise directions, respectively. The origin is located at the upper left corner of the left heater, as shown in the inset of Fig. 1. Details of the layout of the measurement plane locations for the row-averaged (X_i) and column-averaged (Z_i) temperature difference between the local surface temperature and the reference temperature along the rear face of each heater are shown in Fig. 2. Water enters through the upper opening of a gravity-driven vertical square duct of width $H = 50$ mm and 1100 mm long with a uniform velocity u_0 and temperature T_0 . The temperature at the test section inlet is maintained fixed using a constant-temperature refrigerated bath (TECHNE RB5A/TU-20D) with a resolution of ± 0.1 K and temperature stability of ± 0.05 K. The temperature of the water at the inlet of the test section is measured using a high-precision thermocouple probe (Control Company, 4132 Traceable® Platinum RTD Thermometer). Measurements confirmed a constant entry temperature

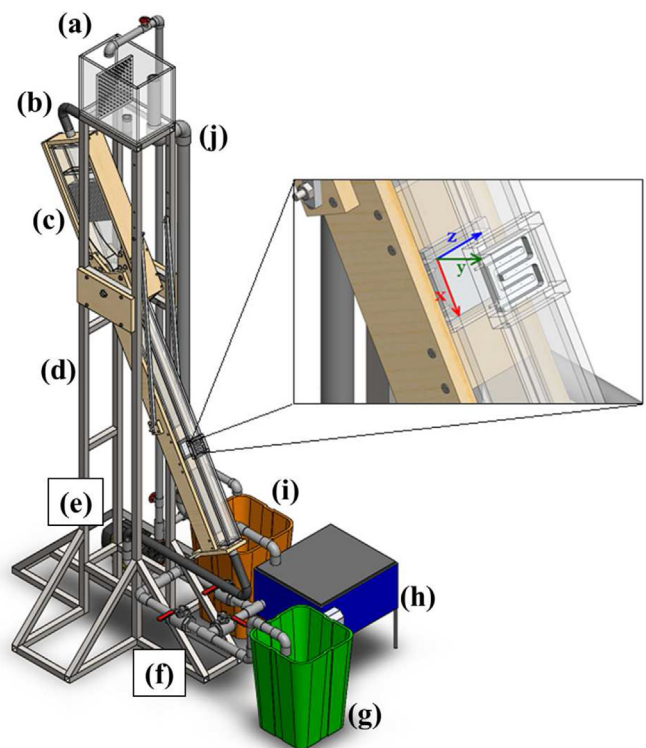


Fig. 1. Schematic diagram of the experimental setup. (a) Constant head tank. (b) Flexible hose. (c) Nozzle section with honeycomb and mesh structures. (d) Metal framework. (e) Centrifugal pump. (f) Adjustable valve. (g) Graduated container. (h) Constant-temperature refrigerated bath. (i) Reservoir tank. (j) Overflow tube.

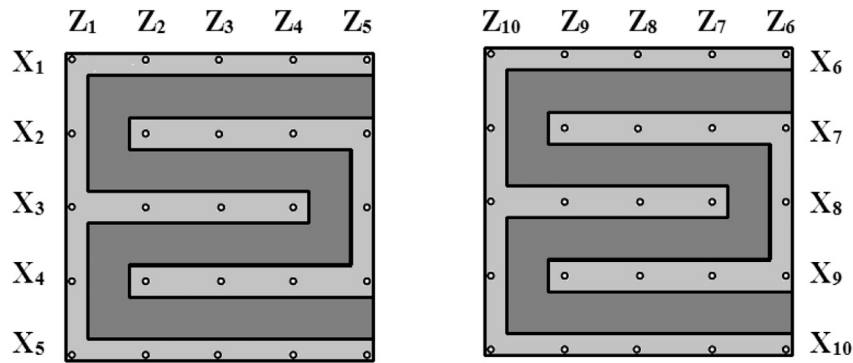


Fig. 2. Schematic layout of the measurement plane locations for the column-averaged (Z_i) and row-averaged (X_i) temperature difference between the local surface temperature and the reference temperature $\Delta T = (T_{wj} - T_0)$ along the rear face of each heater.

of 293 K. A constant-head tank is used to maintain a steady inflow condition during the experiment, and it is filled from a reservoir tank using a centrifugal pump. A section with honeycomb and mesh structures is used prior to the channel entrance to produce a uniform velocity, and it is connected to the constant head tank by means of a flexible hose.

The nozzle has been designed to eliminate flow separation and increase flow alignment, and it is made of 10 mm thick Plexiglas with a contraction ratio of 4:1. In order to qualitatively assess the overall vortex structure, the water in the channel is seeded with neutrally buoyant polyamide spherical particles of 20 μm in diameter. The heaters are located 700 mm downstream of the channel inlet, providing a minimum hydrodynamic entry length of 14 hydraulic diameters. This allows the fluid laminar boundary layer to be fully developed prior to reaching the heaters. The channel walls are 10 mm thick and are made of Plexiglas ($k \approx 0.189 \text{ W/m K}$) with two identical square aluminum ($k \approx 249 \text{ W/m K}$, $\epsilon = 0.09$) flush-mounted, symmetric and discrete heat sources with uniform surface heat flux towards the fluid located at the midheight of opposite sidewalls. Both plates are 10 mm thick, 50 mm long and 50 mm wide, thus occupying the whole channel depth. Each one is heated with constant heat flux using a Nickel–Chromium alloy resistance heating ribbon wire (Omega Engineering, NCR-25-100) that is inserted through a machined groove as shown in Fig. 3.

High thermal conductivity paste (Omega Engineering, OT-201-16) was pressed to fill the gap between the heating ribbon wire and the inner wall of the aluminum plate, and the heating ribbon wire is electrically insulated and firmly bonded with a high thermal conductivity epoxy resin (Omega Engineering, OB-101) that replenishes the slit. In order to minimize the external heat loss, the rear surface of each heater is insulated with a 10 mm thick Plexiglas cover with holes drilled for the thermocouples and an external 25 mm thick polystyrene foam insulation ($k \approx 0.0033 \text{ W/m K}$).

2.2. Heater construction

To produce the uniform heat flux, each heating element is powered by a triple-output 30 V–5 A DC variable voltage–current power supply (BK Precision, 1671A), and power dissipation is determined by measuring their voltage and current drop. Each heater block is instrumented with twenty five equally spaced copper-constantan (T-type, 0.38 mm) thermocouples (Omega Engineering, COCO-015) that are embedded in holes drilled from the back surface of the plate at a depth of 9.8 mm and that are located 11.75 mm apart, as shown in Fig. 4.

Good physical contact between the junction of each thermocouple is achieved using the high thermal conductivity paste, and epoxy resin is applied to the drilled holes to hold the thermocouples in place. Signals from the thermocouples are recorded by a data acquisition system (Omega Engineering, OMB-DAQSCAN-2005) that is equipped with a 56 channel thermocouple input module (Omega Engineering, OMB-DBK90). The *in situ* surface temperature of the heated plates T_w , the fluid's temperature at the channel inlet, and the ambient temperature are measured simultaneously with a sampling frequency of 1 Hz, with a measurement period that varies from 10 to 120 min depending on the value of the Reynolds and

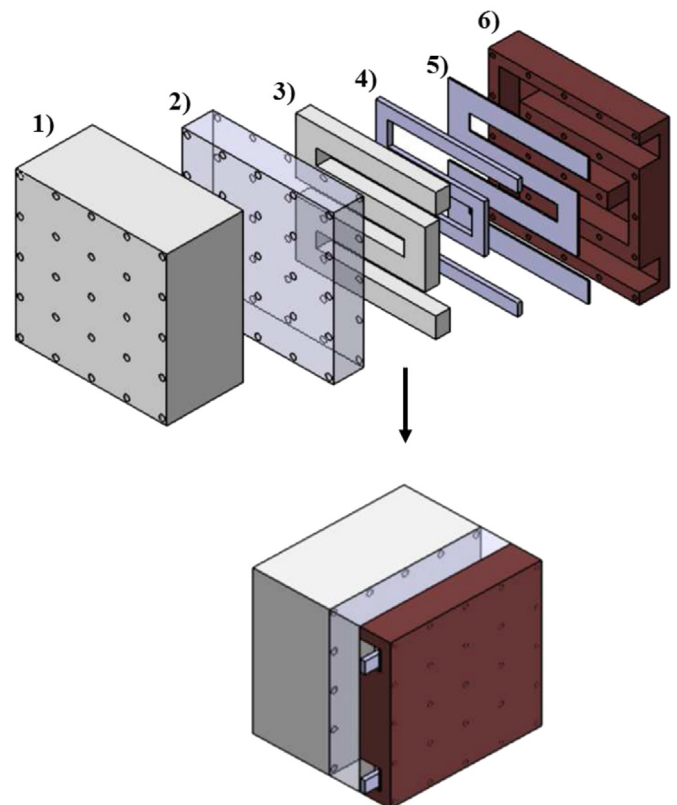


Fig. 3. Disassembled heater (top image) with all its components: 1) Polystyrene foam insulation with holes drilled for the thermocouples. 2) Plexiglas cover with holes drilled for the thermocouples. 3) High thermal conductivity epoxy resin. 4) Nickel–Chromium alloy resistance heating ribbon wire. 5) High thermal conductivity paste. 6) Aluminum heating element. The lower image shows the assembled heating element.

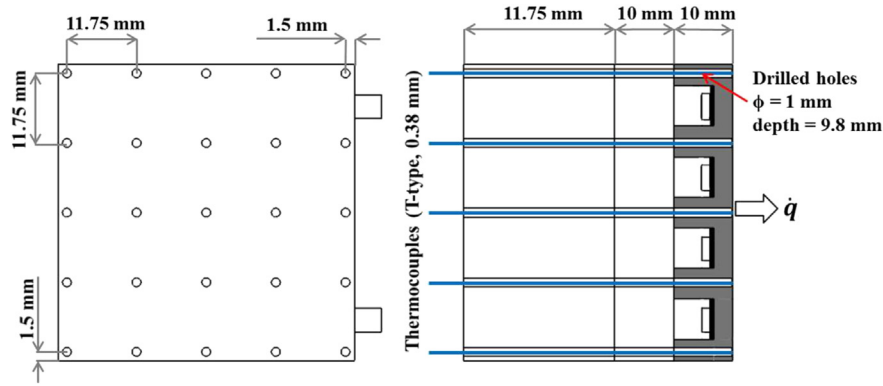


Fig. 4. Dimensions of each discrete heater. The hollow circles illustrate their thermocouple distribution for local heat transfer measurements. Left image: Frontal view facing the fluid. Right image: Lateral view.

modified Richardson numbers. Signals from the thermocouples are collected, processed, stored, and analyzed with the data acquisition system interfaced with a PC. All thermocouples are calibrated with the constant temperature bath and the measurement error is found to be within ± 0.1 K. The measurements are done when the experimental conditions reach a steady-state condition after approximately 1–3 h. To minimize conduction heat losses through the channel walls, a 25 mm thick polystyrene foam insulation is applied to all the channel walls, including the constant-head tank and nozzle. All fluid properties are computed based on the thermophysical properties evaluated with the arithmetic mean between the average temperature of the slabs and the fluid temperature at the inlet [41].

2.3. Parametric set

The set of parameters can be obtained as follows, for a Newtonian Boussinesq fluid downflowing from the entrance of an inclined rectangular channel of width H and inclination angle γ . Both walls have two identical flush-mounted, symmetric and discrete heat sources of length l_2 , with uniform surface heat flux $\dot{q}$ towards the fluid. As the fluid is heated then buoyancy acts against the flow, producing different thermal and flow patterns. The heated surface temperatures can be then obtained, being functions of the heat flux, $\dot{q}$, the fluid thermal conductivity, k , and the characteristic thicknesses of the thermal layers, δ_j , with $j = R, L$ for the right and left heated slabs. That is $T_{wj} - T_0 = \Delta T_j = F_j(\dot{q}, k, \delta_j; t)$, where t denotes the time. The thermal layers δ_j depend on the forced and natural convection flow parameters: inlet velocity u_0 , kinematic viscosity ν , thermal diffusivity α , gravity g , thermal expansion coefficient β , inclination angle γ , characteristic length of the channel D_H and additional geometrical parameters l_i (length of the channel, length and position of the heated slabs). Here, D_H is the hydraulic diameter of the channel (characteristic length) calculated from $D_H = 4A_c/P$, where A_c is the cross-sectional area and P is the channel perimeter. Therefore, the resulting temperature differences $\Delta T_j = F_j(\dot{q}, k, u_0, \nu, \alpha, g, \beta, \gamma, D_H, l_i; t)$. A set of nondimensional parameters can be obtained as

$$\frac{\Delta T_j k}{\dot{q} D_H} = f_j \left(\frac{\nu}{u_0 D_H}, \frac{\alpha}{u_0 D_H}, \frac{g D_H}{u_0^2}, \frac{\beta \dot{q} D_H}{k}, \gamma, \frac{l_i}{D_H}, \frac{u_0 t}{D_H} \right), \quad (1)$$

or employing the well known parameters (using the fact that the third and fourth parameters on the right hand side arise always together)

$$Nu_j = G_j(Re, Pr, Ri^*, \gamma, L_i; \tau), \quad (2)$$

where Nu_j are the Nusselt numbers, $Nu_j = \dot{q} D_H / \Delta T_j k$, Re is the Reynolds number, $Re = u_0 D_H / \nu$, Pr is the Prandtl number, $Pr = \nu / \alpha$, Ri^* is a heat flux modified Richardson number $Ri^* = g \beta \dot{q} D_H^2 / k u_0^2$, $L_i = l_i / D_H$ and the nondimensional time $\tau = u_0 t / D_H$. The Reynolds number is varied by operating an adjustable valve that regulates the volume flow rate. For a given Reynolds number, the Richardson number is modified by varying the net convection heat flux transferred to the fluid through each heater face. When $T_w > T_0$, the flow reverses and complex vortex structures appear close to the heated slabs. The transient local and mean heat transfer characteristics of these structures are reported and described in the following section.

2.4. Data reduction analysis

The net convective heat flux per unit surface transferred to the fluid through each heater face is calculated from the following energy balance

$$\dot{q} = \frac{\dot{Q}}{A_{heater}} = \dot{q}_{el} - \dot{q}_{cond} - \dot{q}_{rad}, \quad (3)$$

where A_{heater} is the surface area of each heater exposed to the water flow. Here, $\dot{q}_{el}$ and $\dot{q}_{rad}$ are the measured input power supplied to each heater and the calculated radiation losses to the ambient dissipated through the insulation on the rear face of each heater, respectively. $\dot{q}_{cond}$ are the calculated conduction losses to the ambient dissipated through the rear and lateral faces of each heater through the insulation. The total heat input supplied to each heater is $\dot{q}_{el} = VI / A_{heater}$ (Joule effect), where V and I are the measured voltage and electric current, respectively. Radiative heat transfer dissipated through the rear face of the heater can be estimated as (gray body radiation in a optically thin gas environment)

$$\dot{q}_{rad} = \sigma \epsilon F (\bar{T}_w^4 - T_{amb}^4) \approx 4 \sigma \epsilon F T_{amb}^3 \Delta T (1 + 3/2 \Delta T / T_{amb} + \dots), \quad (4)$$

Here, $\Delta T = \bar{T}_w - T_{amb}$. Neglecting the second term inside brackets ($\Delta T / T_{amb} \ll 1$), the radiative heat loss approximates to $\dot{q}_{rad} \approx 4 \sigma \epsilon F T_{amb}^3 \Delta T$. In Eq. (4), σ is the Stefan–Boltzmann constant, the view factor F between the heat source and its surroundings is a constant of order unity [42–46], the measured surface emissivity of aluminum is 0.05 with an experimental error of 4%, $\bar{T}_w$ is the mean

wall or surface temperature of each heater obtained with the average of the temperatures measured by the twenty five thermocouples, and T_{amb} is the ambient temperature. From Eq. (3), the ratio $\dot{q}/\dot{q}_{el}$ can be shown to be given by

$$\frac{\dot{q}}{\dot{q}_{el}} = 1 - \frac{1}{(Nu)_{el}}(r_{cond} + r_{rad}), \quad (5)$$

where $(Nu)_{el} = \dot{q}_{el}D_H/(k(\bar{T}_w - T_0))$, $r_{cond} = (k_l D_H/(k d_l))$ and $r_{rad} = 4\sigma\epsilon F T_0^3 D_H/k$. Here k_l and d_l are the thermal conductivity and the mean thickness of the insulator. In the present case, with the adequate thermal insulation described in Section 2.2, the following values are obtained for the heat loss coefficients: $r_{cond} \approx 0.02$ and $r_{rad} \approx 0.05$. With a typical value employed in this work of $(Nu)_{el}$ of 30 (see Section 3), then $\dot{q}/\dot{q}_{el} = Nu/(Nu)_{el} \approx 0.997$. Therefore, the use of $\dot{q}_{el}$ instead of $\dot{q}$ represents an error below 1%.

The local Nusselt numbers Nu_j can be then obtained as

$$Nu_j = \frac{\dot{q}D_H}{k(\bar{T}_{wj} - T_0)}. \quad (6)$$

Based on the local surface temperature signals recorded by the thermocouples, the time-variations of the local Nusselt numbers along the spanwise and axial distance for each heater can be obtained. The space average value of the Nusselt number $(\bar{Nu})_j$ for the left and right heaters is calculated based on their calculated average surface temperature $\bar{T}_w$ as follows:

$$\bar{Nu}_j = \frac{\dot{q}D_H}{k(\bar{T}_w - T_0)}. \quad (7)$$

For each value of the Reynolds number, a total of three runs have been conducted for each value of Ri^* and inclination angle γ . In order to avoid the possibility of hysteresis effects that could be present depending upon the flow's history [47], for all cases, the measured flow and thermal patterns are achieved by adding opposing buoyancy to a forced convection flow ($Ri^* = 0$) by monotonically increasing the value of the modified Richardson number. To remove the high frequency noise from the time series data, a filtering method based on adjacent averaging smoothing has been used.

2.5. Uncertainty analysis

An estimation of the overall uncertainty in the experimental data is made using standard techniques for single-sample measurements of Kline and McClintock [48]. The uncertainty of a dependent variable R as a function of the uncertainties in the independent variables $x_1, x_2, \dots, x_N$, is given by the relation

$$R = R(x_1, x_2, \dots, x_N), \quad (8)$$

The calculated total uncertainty is obtained from the following equation,

$$\delta R = \sqrt{\sum_{i=1}^N \left(\frac{\partial R}{\partial x_i} \delta x_i \right)^2}. \quad (9)$$

where δx_i are the uncertainties in the measured variables x_i . Our analysis indicates that the experimental uncertainties at a 95% confidence level for the Reynolds, Richardson and Nusselt numbers are 5.9%, 5.1% and 6.2%, respectively. These values are based on the assumption of negligible uncertainty in the relevant fluid properties.

3. Experimental results and analyses

In this section, results of the present investigation are presented and discussed. To illustrate the effects of the duct inclination on the overall nondimensional heat transfer of each heater, results for various inclination angles are presented for fixed Reynolds number of $Re = 700$ under varying thermal buoyancy. Except for the horizontal configuration ($\gamma = 0^\circ$) where a maximum value of $Ri^* = 127.48$ was reached, the maximum value of the modified Richardson number in this study was limited to $Ri^* = 93.02$, as for higher values of buoyancy strength, the reversed flow climbed to upstream regions inside the channel and reached positions that were close to the channel inlet.

3.1. Vertical channel, $Re = 700$, $\gamma = 90^\circ$

For the vertical channel, experimental results for the space-averaged heat transfer characteristics show that for relatively small values of the modified Richardson number ($Ri^* \approx 31.19$), flow reversal takes place close to each heater and a pair of thin recirculation bubbles develop due to baroclinity. Because there is only a weak interaction between both recirculation bubbles, the measured space-averaged Nusselt numbers ($\bar{Nu}$) practically (not really) reach a symmetric steady-state, with a mean $\pm$ standard deviation of 26.52 ± 0.52 and 26.65 ± 0.53 at the left and right heater, respectively. For $Ri^* = 54.74$, a stable nonsymmetric heat transfer pattern takes place and vortex migration to higher positions inside the channel occurs. As a result, higher heat transfer rates are obtained, and the measured spaced-averaged Nusselt numbers at the left and right heaters reach 32.21 ± 0.41 and 34.46 ± 0.44 , respectively. For $Ri^* = 93.02$, flow asymmetry increases, heat transfer rates step up and the measured temporal evolution of the space-averaged Nusselt numbers at the left and right heated surfaces reach 38.43 ± 0.32 and 41.44 ± 0.37 , respectively. Nonetheless, for this value of the buoyancy parameter, a steady-state nonsymmetric heat transfer pattern sets in, indicating that for this channel configuration and for the range of modified Richardson numbers studied, the threshold value for the transition from steady to time-periodic flow and heat transfer is not reached. Fig. 5 shows for selected runs the heaters temperature difference distribution $(T_w(X,Z) - T_0)$ during steady state heat transfer for $Ri^* = 31.19$, 54.74 and 93.02. For comparison purposes, the measured values in Fig. 5 are plotted against those obtained by Barreto et al. [49], who performed a similar experimental study using the same Reynolds and Prandtl numbers and where the only difference between their work and the present one is that the symmetric discrete heat sources subjected to a constant wall heat flux boundary condition partially block the channel instead of being flush-mounted. Note that this is in a sense an unfair comparison, as the modified Richardson numbers employed are not the same. However, it is useful for assessment of the heat transfer characteristics. In this figure, the closed symbols correspond to the mean values obtained with flush-mounted discrete heat sources, and the open symbols correspond to those obtained for each heater block of the partial blockage [49]. Small but detectable heater temperature variation has been measured. Here, the five nondimensional spanwise (Z) and axial (X) distances correspond to $Z_i = X_i = 0.03$, 0.265, 0.5, 0.735 and 0.97. Figs. 5a and b illustrate how the surface temperature varies with spanwise distance. Here, because the recorded surface temperatures are higher at the centerline of the spanwise distance ($Z = 0.5$), the heat flux towards the fluid is lower in the middle of the channel. These results provide powerful evidence that even during steady state heat transfer, because of the presence of wall effects, the flow and heat transfer behavior presents important 3D effects. Interestingly, these images clearly show

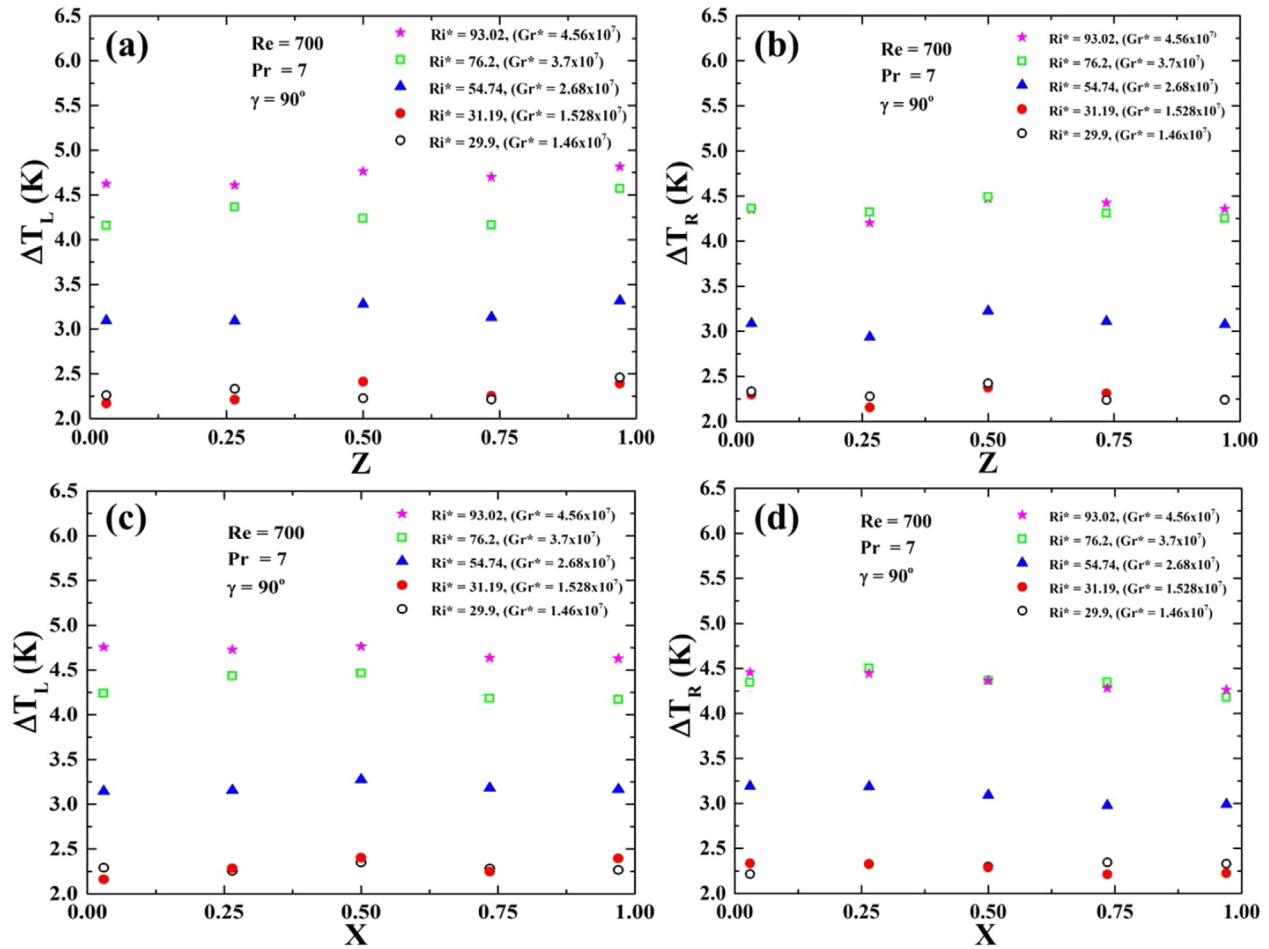


Fig. 5. $Re = 700$, $\gamma = 90^\circ$ and $Ri^* = 31.19, 54.74$ and 93.02 : (a) and (b) Mean $\pm$ standard deviation of the column-averaged (Z_i) temperature difference between the local surface temperature and the reference temperature $\Delta T = (T_{wj} - T_0)$ along the nondimensional spanwise distance during steady state heat transfer at the left (ΔT_L) and right (ΔT_R) heater, respectively. (c) and (d) Mean $\pm$ standard deviation of the row-averaged (X_i) temperature difference $\Delta T = (T_{wj} - T_0)$ along the nondimensional axial distance during steady state heat transfer at the left (ΔT_L) and right (ΔT_R) heater, respectively. The closed symbols correspond to the mean values obtained with flush-mounted discrete heat sources, and the open symbols correspond to those obtained for each heater block of the partial blockage [49].

that the spanwise surface temperature variations have a more important contribution than the axial surface temperature variations in the overall nonsymmetric flow and heat transfer response achieved, and that these variations increase for increasing values of the modified Richardson number. Figs. 5c and d show the effect of Ri^* number variation on the heaters temperature difference distribution with X . Clearly, a more uniform surface temperature takes place because of the presence of secondary flow.

3.2. $Re = 700$, $\gamma = 60^\circ$

For all inclined channel configurations, except for the vertical one, a nonsymmetric Nusselt number distribution takes place for all non-zero values of the buoyancy parameter. For a modified Richardson number of $Ri^* = 31.19$, a dramatic increase in the heat transfer rate with respect to the purely forced convection flow ($Ri^* = 0$) takes place, a stable nonsymmetric thermal response is reached, and the measured spaced-averaged Nusselt numbers at the left and right heaters reach 30.7 ± 0.74 and 29.7 ± 0.61 , respectively. For a higher value of the modified Richardson number ($Ri^* = 54.74$), a stationary and nonsymmetric thermal response is still reached and the measured spaced-averaged Nusselt numbers at the left and right heaters increase to

32.5 ± 0.94 and 34.4 ± 1.24 , respectively. As the value of the modified Richardson number increases to $Ri^* = 93.02$, transition from steady to time-periodic flow and heat transfer takes place. For this channel configuration a large right/upper recirculation bubble located next to its corresponding heater and occupying upstream regions of the channel is present, while a smaller left/lower recirculation bubble is observed next to its corresponding heater and downstream of the latter. Figs. 6a and b show the effect of the channel orientation on the temporal evolution of the space-averaged temperature difference distribution and space-averaged Nusselt number of each heater for $\gamma = 60^\circ$ and $Ri^* = 93.02$, respectively. From these figures, it is clear that quasi-periodic flow and heat transfer sets in. The mean surface temperature of the left heater is slightly lower than the right one, while the opposite occurs for the overall Nusselt number. The measured overall Nusselt numbers for the left and right heaters are 38.5 ± 0.69 and 37.3 ± 0.59 , respectively. The relatively small values of the standard deviation of both signals indicates that the temperature oscillations that are present are of small amplitude. The normalized power spectra of the oscillations of the space-averaged Nusselt numbers for $Ri^* = 93.02$ are shown in Fig. 6c as functions of the nondimensional oscillation frequency or Strouhal number, $St = fD_H/u_0$. Notice how a periodic heat transfer response with

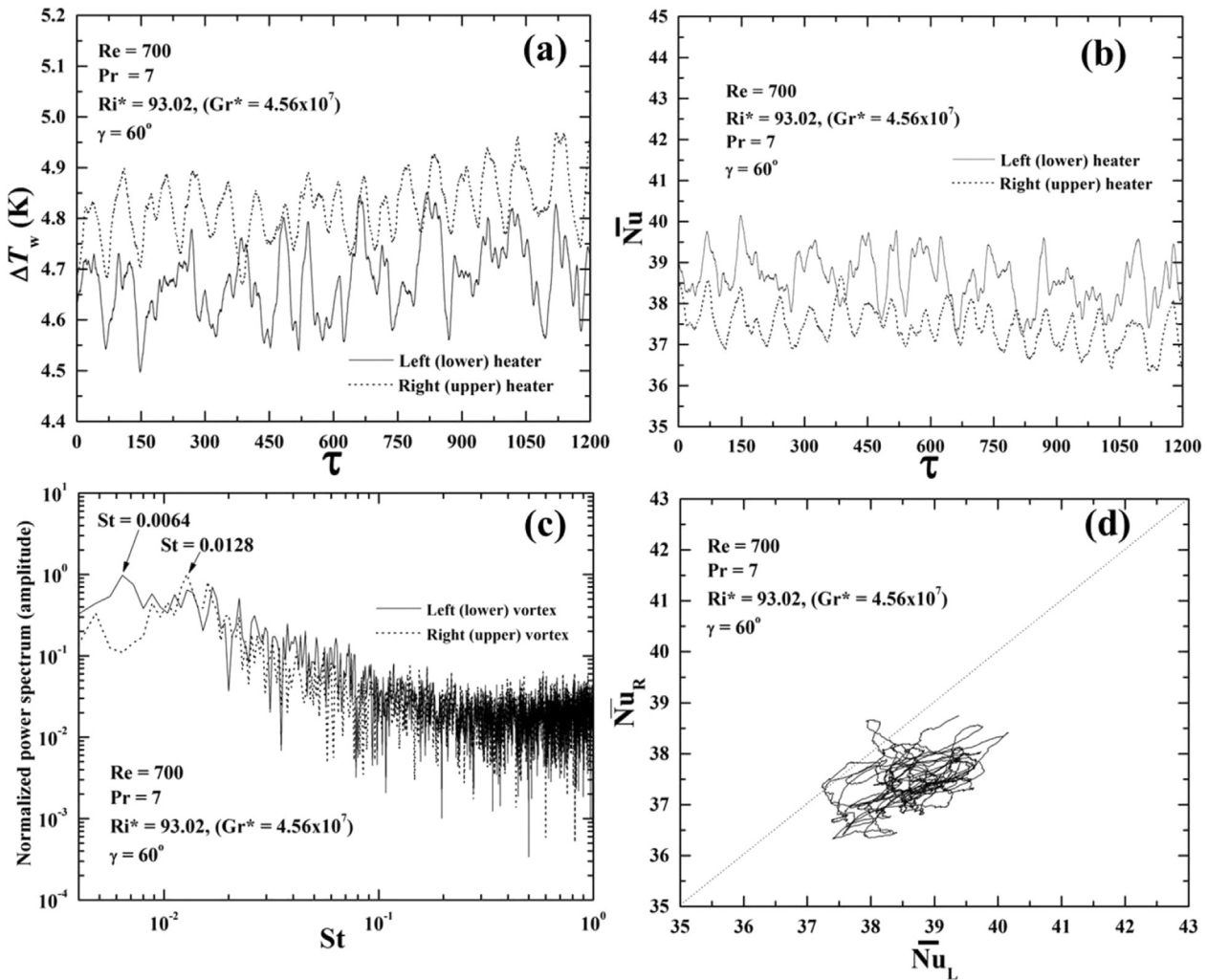


Fig. 6. $Re = 700$, $Ri^* = 93.02$ and $\gamma = 60^\circ$: (a) Temporal evolution of the space-averaged temperature difference between the mean surface temperature and the reference temperature $\Delta T = (\bar{T}_{wj} - T_0)$ of each heater. (b) Temporal evolution of the space-averaged Nusselt number of each heater. (c) Normalized power spectra of the space-averaged Nusselt numbers. (d) Phase-space plot of the space-averaged Nusselt number at the left heater ($\bar{Nu}_L$) as a function of the averaged Nusselt number at the right heater ($\bar{Nu}_R$).

double frequency is achieved with a multi-spectral behavior with several peaks, ranging from $St \sim 0.0064$ (lower vortex) to $St \sim 0.0128$ (upper vortex). Fig. 6d shows for $Ri^* = 93.02$ the trajectories in the phase-space plot of the space-averaged Nusselt number at the left heater ($\bar{Nu}_L$) as a function of the space-averaged Nusselt number at the right heater ($\bar{Nu}_R$), illustrating how a time-periodic flow and heat transfer takes place. Fig. 7 shows the time-variations of the heaters temperature difference along the spanwise and axial distance of each heat source for $Ri^* = 93.02$.

Figs. 7a and b illustrate the presence of spanwise non-uniformity at the left and right heaters, respectively. Here, the recorded transient asymmetric temperature distribution of each heater reveals that the heat fluxes and thus the local column-averaged Nusselt numbers at the Z_1 and Z_2 positions of the left heater and Z_6 and Z_7 positions of the right heater are higher. Interestingly, note how the lowest heat fluxes take place at the centerline of their corresponding heaters (Z_3 and Z_8), indicating that the highest flow reversal takes place close to the channel corners. The temperature difference distribution for the left and right heaters are shown in Figs. 7c and d, respectively. From these figures, it is apparent that because of the presence of a relatively small recirculation bubble next to the left heat source, the latter

achieves a more homogeneous surface temperature. As can be seen from Fig. 7d, because the core of the right oscillating vortex is located upstream of its corresponding heat source, the heat flux increases towards the axial direction and the amplitude of their oscillations decreases.

3.3. $Re = 700$, $\gamma = 45^\circ$

For this channel inclination, experimental measurements show that steady-state heat transfer is achieved for modified Richardson numbers up to $Ri^* = 31.19$. Fig. 8 shows for selected runs the influence of the buoyancy parameter for $\gamma = 45^\circ$ by displaying results up to a final nondimensional time of $\tau = 1000$. Fig. 8a shows the temporal evolution of the space-averaged Nusselt number of each heater for $Ri^* = 54.74$ and 93.02 . For these values of the buoyancy parameter, the Hopf bifurcation is triggered and an oscillatory flow and heat transfer sets in. For $Ri^* = 54.74$, a large vortex structure that extends from the right heater to upstream positions inside the channel is present at the right wall, while a smaller elongated vortex located close to the left heater and that extends towards the downstream direction is observed. For this flow structure, the overall Nusselt number of the right heater is always higher than the left one. Nevertheless, both signals oscillate with a relatively small

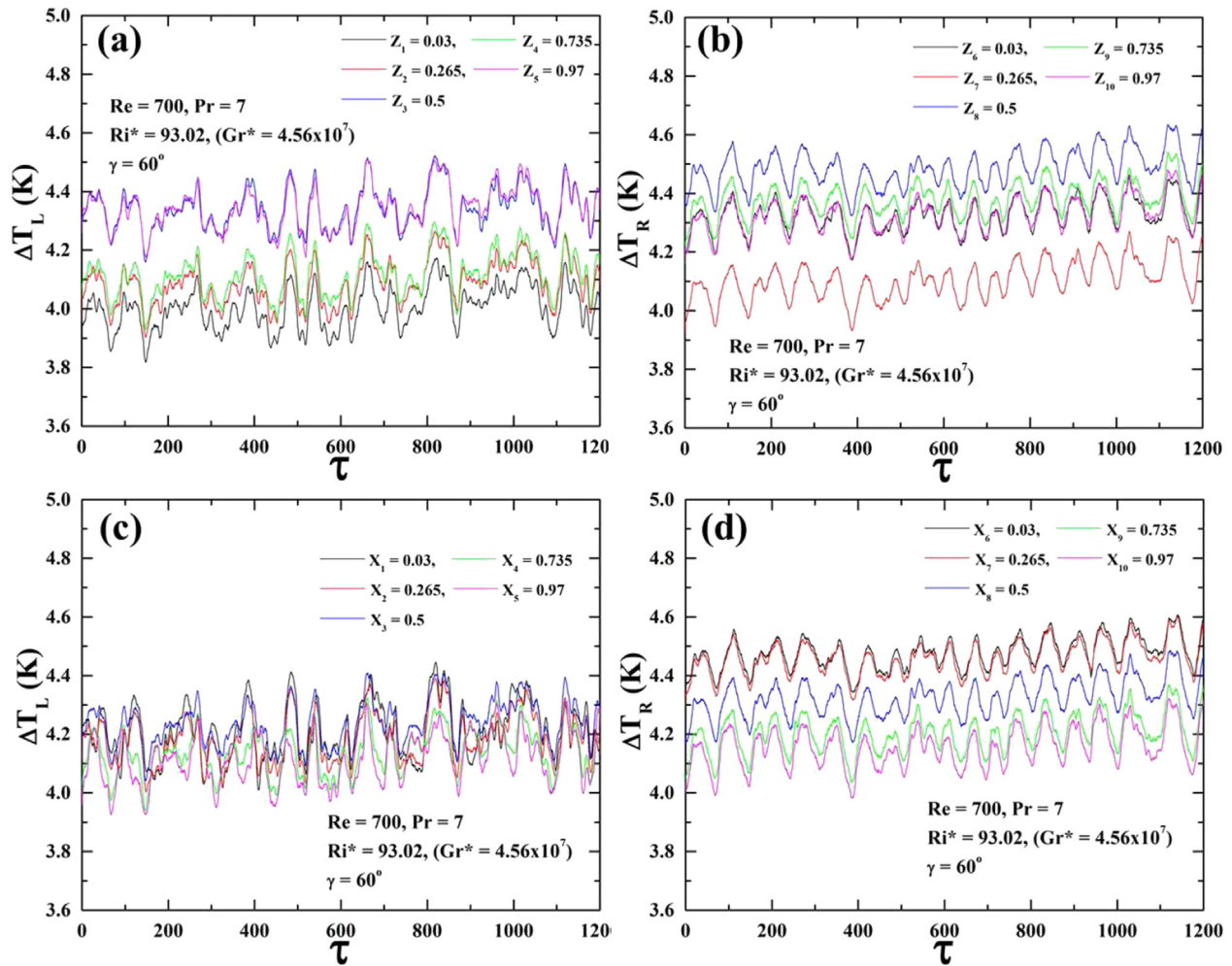


Fig. 7. Time-variations of the column-averaged (Z_i) and row-averaged (X_i) temperature difference between the local surface temperature and the reference temperature $\Delta T = (T_{w,j} - T_0)$ along the nondimensional spanwise and axial distance at $Re = 700$, $\gamma = 60^\circ$ and $Ri^* = 93.02$.

amplitude and with the same phase. For a higher value of the buoyancy parameter ($Ri^* = 93.02$), the size and strength of both vortex structures increases, the amplitude of the oscillations of the space-averaged Nusselt number of each heater increase and their frequency steps up. Interestingly, a phase-lag of π rad. appears between both signals, indicating that the oscillating 3D vortex structure fluctuates in the longitudinal and spanwise direction (flapping). Fig. 8b shows the trajectories in the phase-space plot of the space-averaged Nusselt number of both heaters for $Ri^* = 54.74$ and 93.02 , illustrating the periodic behavior of both signals. For comparison purposes, Figs. 8a and b show the measured values obtained for the discrete heater blocks described in Ref. [49] for a modified Richardson number of $Ri^* = 76.2$.

For this value of buoyancy strength, the space-averaged Nusselt numbers of the heater blocks on the channel floor and roof are slightly higher and about equal than those obtained with the flush-mounted heated slabs for $Ri^* = 54.74$, respectively. These results reveal that because the discrete heat sources are flush-mounted, the persistence of a large single-core vortex structure on the channel roof allows for the development of higher flow reversal next to its corresponding heated slab, which results in higher heat transfer rates. Conversely, for the case of the partial blockage, a change in the vortex dynamics of the vortical structures takes place, with interesting splitting and merging between the three vortices on the channel roof. In this context, lower heat transfer

rates are achieved because of the reduced strength of the smaller vortices. Making reference to the space-averaged Nusselt numbers on the channel floor, because for both channel geometries the vortical structure with several pairing vortices of smaller size is found to persist, the heat transfer rates achieved (for different values of buoyancy strength) are about the same. The normalized power spectra of the overall Nusselt numbers for $Ri^* = 54.74$ and 93.02 are shown in Figs. 8c and d, respectively. Clearly, each one has a peak at the corresponding Strouhal number of $St = 0.00861$ and $St = 0.01052$ (time period close to 116 and 95 nondimensional time units), respectively. The time-variations of the temperature differences along the spanwise and axial distance of each heater are shown in Fig. 9 for a modified Richardson number of $Ri^* = 93.02$. In Figs. 9a and b, it can be appreciated that for both heaters, the temperatures are higher at the middle spanwise positions of both heaters (Z_3 and Z_8) than those registered at the other four spanwise locations. Also, it is to be noticed how the spanwise variations and amplitude of the temperature of the left heater are larger than those observed for the right one. These are justified by the presence of a left and relatively small vortex next to its corresponding heater that exhibits strong flow oscillations, while the larger right vortex located at upstream positions of its corresponding heater exhibits weaker oscillations. The temperature difference distribution along the axial distance for the left and right heaters is shown in Figs. 9c and d, respectively. Although they

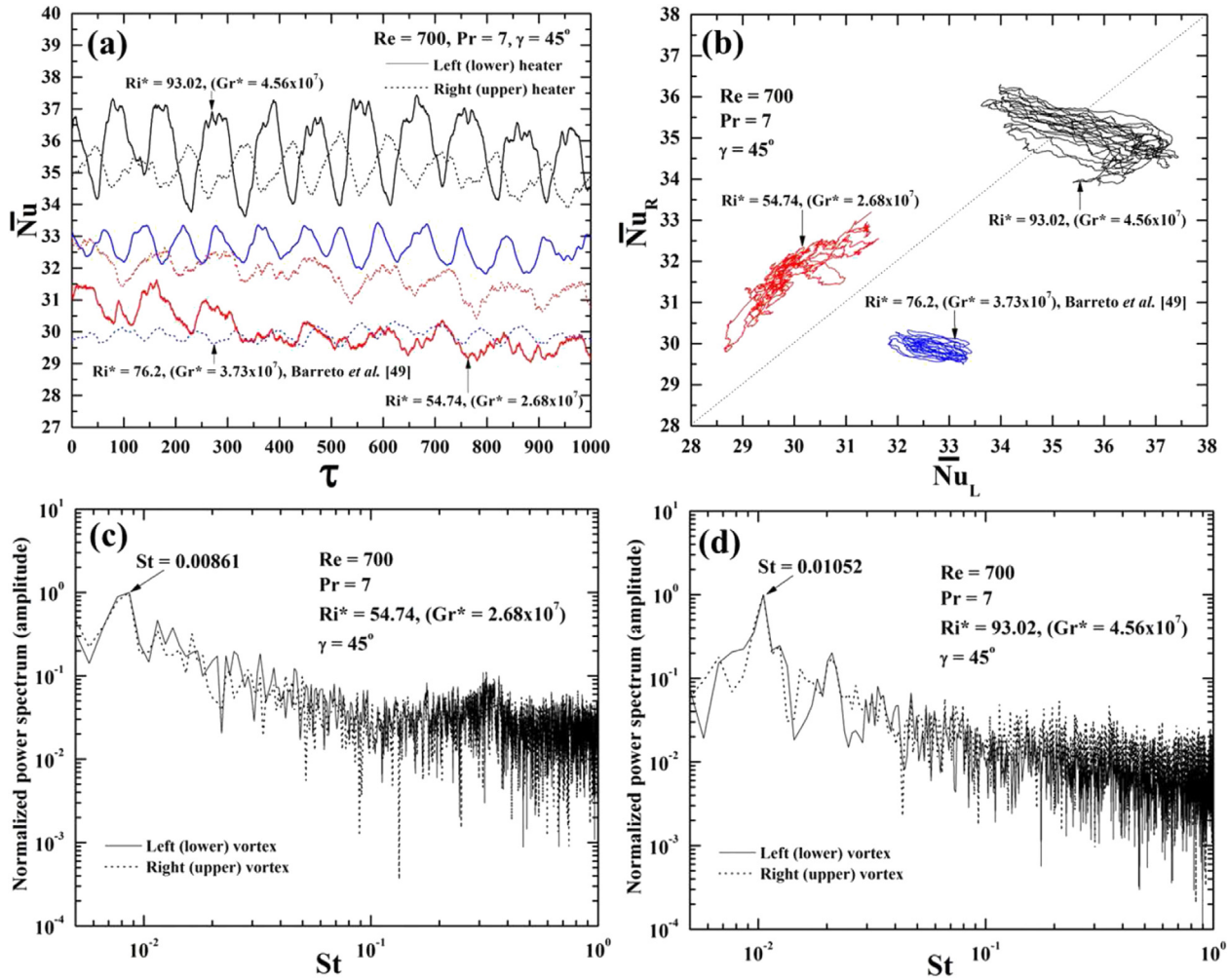


Fig. 8. $Re = 700$ and $\gamma = 45^\circ$: (a) Temporal evolution of the space-averaged Nusselt number of each heater for $Ri^* = 54.74$ and 93.02 . (b) Phase-space plot of the space-averaged Nusselt number at the left heater ($\bar{Nu}_L$) as a function of the space-averaged Nusselt number at the right heater ($\bar{Nu}_R$) for $Ri^* = 54.74$ and 93.02 . (c) and (d) Normalized power spectra of the overall Nusselt numbers for $Ri^* = 54.74$ and 93.02 , respectively.

are similar at each time step, a more homogeneous surface temperature and thus the local row-averaged Nusselt number distribution is observed at the left heater, while those for the right heater increase towards the axial direction. This happens because of the 3D nonsymmetric configuration achieved by the secondary vortex flow structures.

3.4. $Re = 700$, $\gamma = 30^\circ$

For this channel inclination, experimental measurements show that steady-state heat transfer takes place at low opposing buoyancy ($Ri^* \leq 54.74$). However, as the value of the buoyancy parameter is further increased, experimental measurements clearly show that after the onset of a Hopf bifurcation, periodic flow oscillation with a well defined frequency is exhibited by the space-averaged Nusselt numbers of both heaters for relatively large modified Richardson numbers, such as $Ri^* = 93.02$.

The fluctuating thermal response for $\gamma = 30^\circ$ is depicted in Fig. 10a by displaying six periods of oscillation of the measured temporal variations in the space-averaged Nusselt number of both heaters up to a nondimensional time of 800. Note how for this parameter values, both signals have a phase-lag of π rad. For comparison purposes, the space-averaged Nusselt numbers and

their corresponding phase-space plot at $Ri^* = 128.6$ for the discrete heater blocks described in Ref. [49] are also plotted in Figs. 10a and b, respectively. Clearly, the heat transfer performance on the channel roof and floor of the flush-mounted heaters is notably more uniform, as the man values of the overall Nusselt numbers are about the same. In contrast, because of the channel constriction, the heat transfer performance of both heater blocks is clearly unequal. Also, due to the smaller passage, the amplitude of the oscillations of the overall Nusselt numbers reduces with respect to those observed for a lower value of Ri^* . Figs. 10b and c show the trajectories in the phase-space plot of the space-averaged Nusselt numbers and the normalized power spectra of the space-averaged Nusselt number of both heaters, respectively. Clearly, a sharp peak of the spectral distribution of the fluctuating energy at $St = 0.008$ (time period close to 125 nondimensional time units) is the manifestation of the low frequency flow and thermal oscillations. The time-variations of the temperature differences along the spanwise and axial distance of each heater are shown in Fig. 11 for a modified Richardson number of $Ri^* = 93.02$.

Here, the amplitude of the temperature oscillations of the left heater are larger than those for the right one, indicating that the spanwise and axial oscillations of the lower left vortex are stronger than those presented by the upper right vortex. In addition, for this

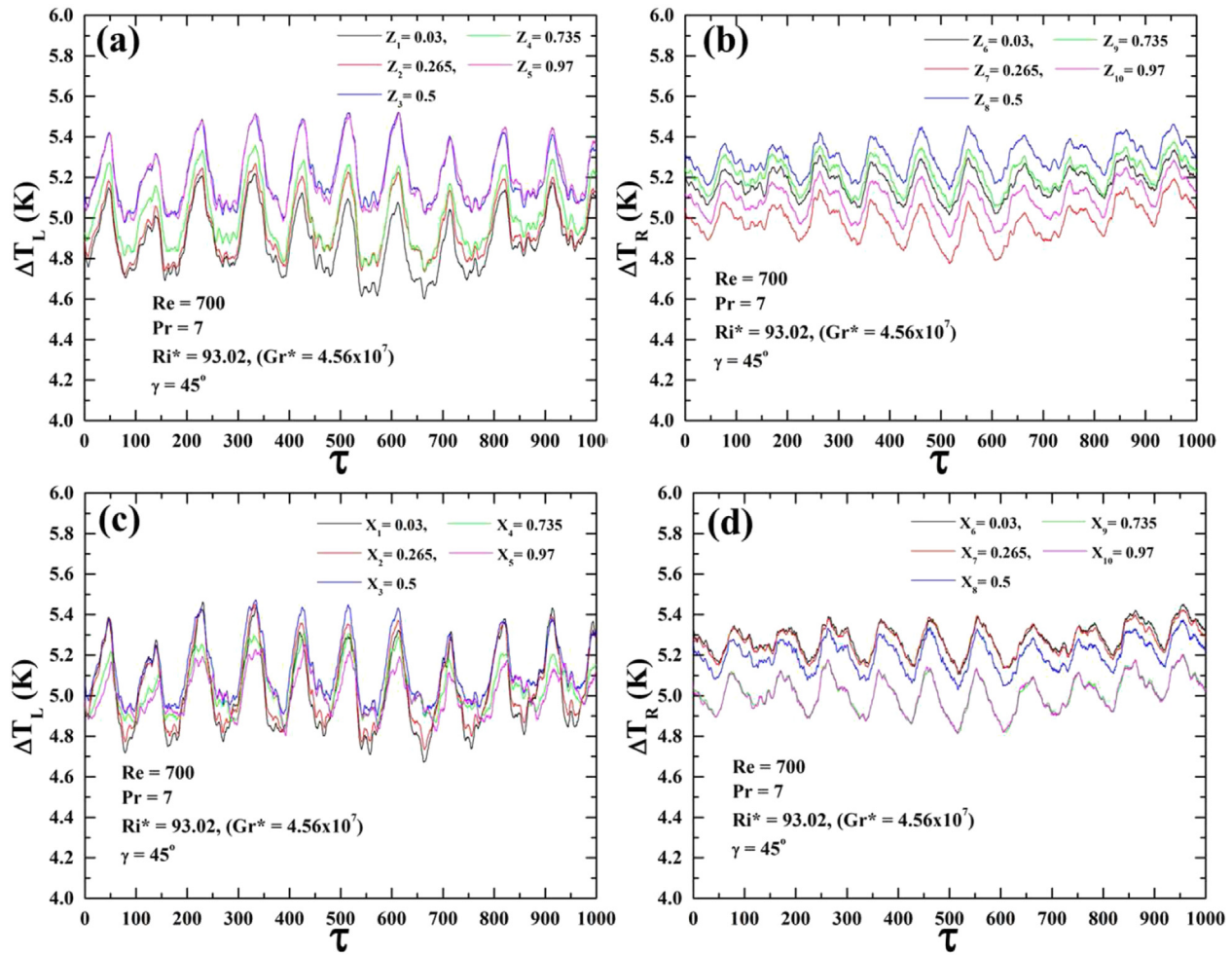


Fig. 9. Time-variations of the measured column-averaged (Z_i) and row-averaged (X_i) temperature difference between the local surface temperature and the reference temperature $\Delta T = (T_{wj} - T_0)$ along the spanwise and axial distance for the left (ΔT_L) and right (ΔT_R) heaters at $Re = 700$, $\gamma = 45^\circ$ and $Ri^* = 93.02$.

channel configuration, the highest heater temperatures are observed at the middle spanwise positions of both heat sources.

3.5. Horizontal channel, $Re = 700$, $\gamma = 0^\circ$

The horizontal channel ($\gamma = 0^\circ$) is the only case where gravity acts only indirectly against the forced flow. As a result, for all the values of the buoyancy parameter studied, steady-state heat transfer is observed. For a pure forced convection flow in the absence of buoyancy ($Ri^* = 0$), the space-averaged Nusselt numbers of both heaters are identical and the flow and thermal response is symmetric. However, as buoyancy sets in, an asymmetric flow and thermal response takes place. For a modified Richardson number of $Ri^* = 31.19$, the value of the space-averaged surface temperature at the bottom heater is lower than the upper one. Hence, the opposite happens with the value of the space-averaged Nusselt numbers. The experimental measurements show that for increasing values of the buoyancy parameter up to $Ri^* = 127.48$, the asymmetry of the thermal response increases monotonically as the Richardson number steps up.

A universal plot with the effect of the channel orientation on the measured time-and-space-averaged Nusselt number ($\overline{Nu}_j = 1/\Delta\tau \int_{\tau_0}^{\Delta\tau} \overline{Nu}_j d\tau$) and the mean convection heat transfer coefficient at each heater for a Reynolds number of $Re = 700$ and several values of the modified Richardson number is shown in

Fig. 12. In this figure, the closed/open symbols correspond to the left (lower)/right (upper) heaters, respectively. Interestingly, for a fixed value of opposing buoyancy, there is not a linear dependence between the obtained value of the mean Nusselt numbers and the inclination angle with respect to the horizontal, indicating that the heat transfer response of the overall Nusselt number against the duct orientation is not monotonic. Clearly, due to the development of buoyancy induced secondary flow acting directly against convective flow, the highest heat transfer rates are obtained for the vertical channel configuration ($\gamma = 90^\circ$). For an inclination angle of $\gamma = 60^\circ$ and a relative small value of the modified Richardson number ($Ri^* = 31.19$), a big jump in the heat transfer rates of both heaters takes place with respect to a purely forced convection flow ($Ri^* = 0$). However, the overall value of the Nusselt numbers illustrate how only small heat transfer enhancement takes place for increasing values of the buoyancy parameter. Also, the asymmetry between the achieved overall Nusselt numbers of both heaters remains practically fixed.

For inclination angles of $\gamma = 45^\circ$ and 30° , the asymmetry in the final flow and heat transfer response reduces for increasing buoyancy. However, this trend is reversed for the horizontal configuration ($\gamma = 0^\circ$). Here, because gravity acts with an opposite effect on both thermal layers, as the upper recirculation bubble increases in size for increasing value of the modified Richardson number, the overall Nusselt numbers do not change much. However, an

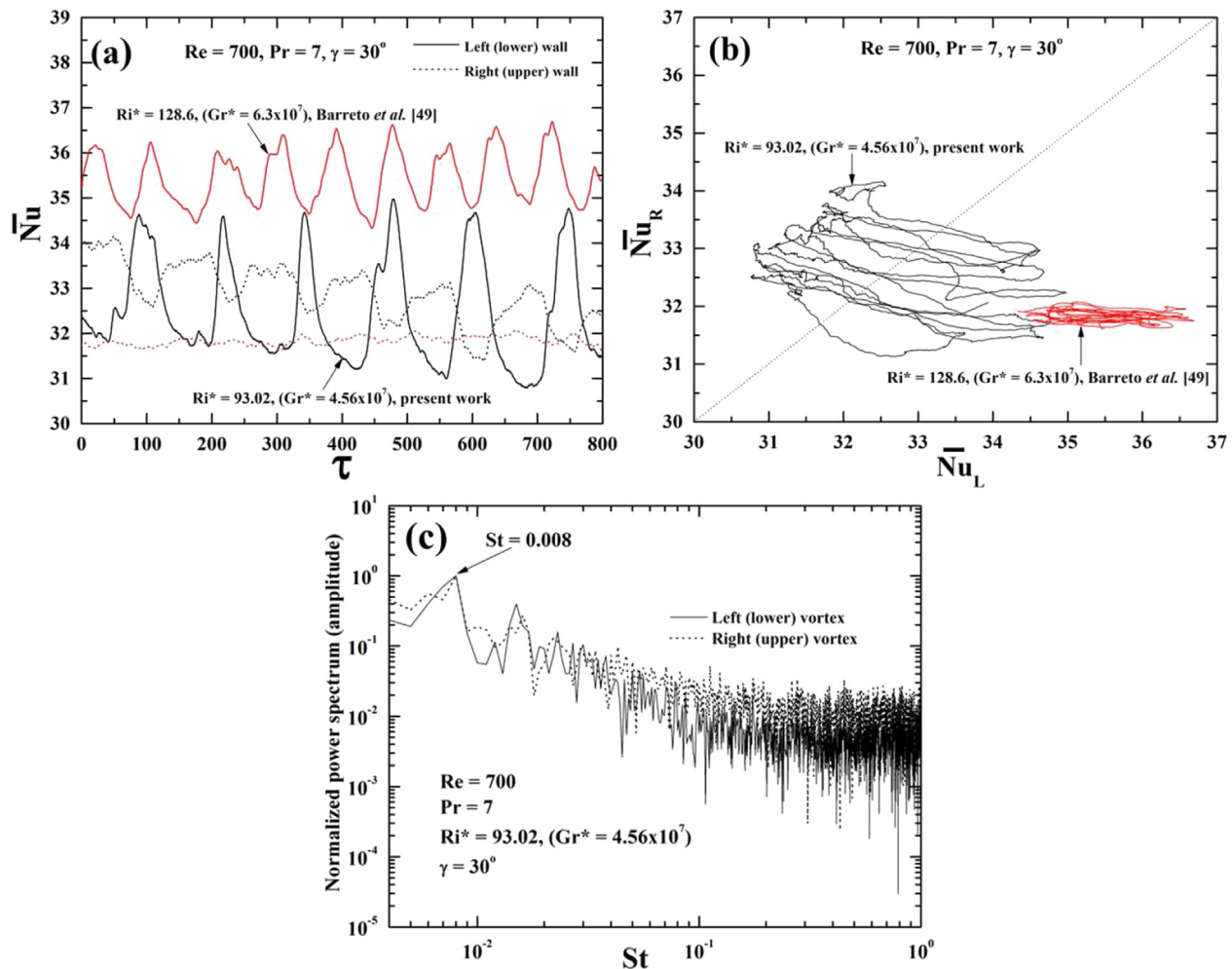


Fig. 10. $Re = 700$, $Ri^* = 93.02$ and $\gamma = 30^\circ$: (a) Temporal evolution of the space-averaged Nusselt numbers. (b) Phase-space plot of the space-averaged Nusselt number of the left heater ($\overline{Nu}_L$) as a function of the space-averaged Nusselt number of the right heater ($\overline{Nu}_R$). (c) Normalized power spectra of the space-averaged Nusselt numbers for $Ri^* = 93.02$.

opposite effect occurs at the lower thermal layer and enhanced heat transfer takes place for increasing values of the buoyancy parameter. Note how for the horizontal configuration, a steady state heat transfer response is observed for modified Richardson numbers up to $Ri^* = 127.48$, indicating that for this value of the inclination angle, the threshold of the buoyancy induced onset of the vortex flow is only reached for relatively high values of the buoyancy parameter. Table 1 shows for $Re = 700$ the required mean net convection heat flux per unit area transferred to the fluid for all the computed values of the modified Richardson number shown in Fig. 12.

3.6. Reynolds number effect

In the following subsection, experimental results of the time- and space-averaged heat transfer characteristics are presented for the vertical channel configuration ($\gamma = 90^\circ$) for Reynolds numbers, $Re = 300, 500, 700$ and 900 and Richardson numbers up to $Ri^* = 93.02$. Figs. 13a and b show for selected runs the measured mean wall-to-coolant temperature difference at each heat source $\Delta T = (\overline{T}_w - T_0)$ and the value of the mean convective heat flux supplied to each heater against Re and three values of the modified Richardson number, respectively. As can be seen from this figure, for a given Reynolds number, the temperature difference and the mean wall heat flux increase monotonically as the modified Richardson number steps up. It is to be noticed that although the

convective heat flux supplied to each heater is equal, because of the nonsymmetric flow and thermal response achieved, unequal values of the heaters' time- and space-averaged surface temperature are registered. Fig. 14 presents both the variations of the time- and space-averaged Nusselt number and the variations of the time- and space-averaged convection heat transfer coefficient at each heater for several values of the Reynolds and modified Richardson numbers. In these figures, it is clear how for a fixed modified Richardson number, heat transfer enhancement is observed for increasing values of the Reynolds number. Also, because of the opposing buoyancy, an asymmetric flow and thermal configuration is observed for all the values of the Reynolds number studied.

4. Conclusions

The problem of steady and oscillatory laminar opposing mixed convection heat transfer in a rectangular channel with two discrete symmetrically, flush-mounted heat sources simulating electronic components has been studied experimentally. The boundary conditions on the heated channel walls are isoflux conditions, and the effects of channel inclination, Reynolds and modified Richardson numbers on the 3D strength and extent of the reversed flow and nondimensional heat flux at each heat source have been presented. From experimental measurements, data for the time variations of the space-averaged surface temperatures, convective heat fluxes

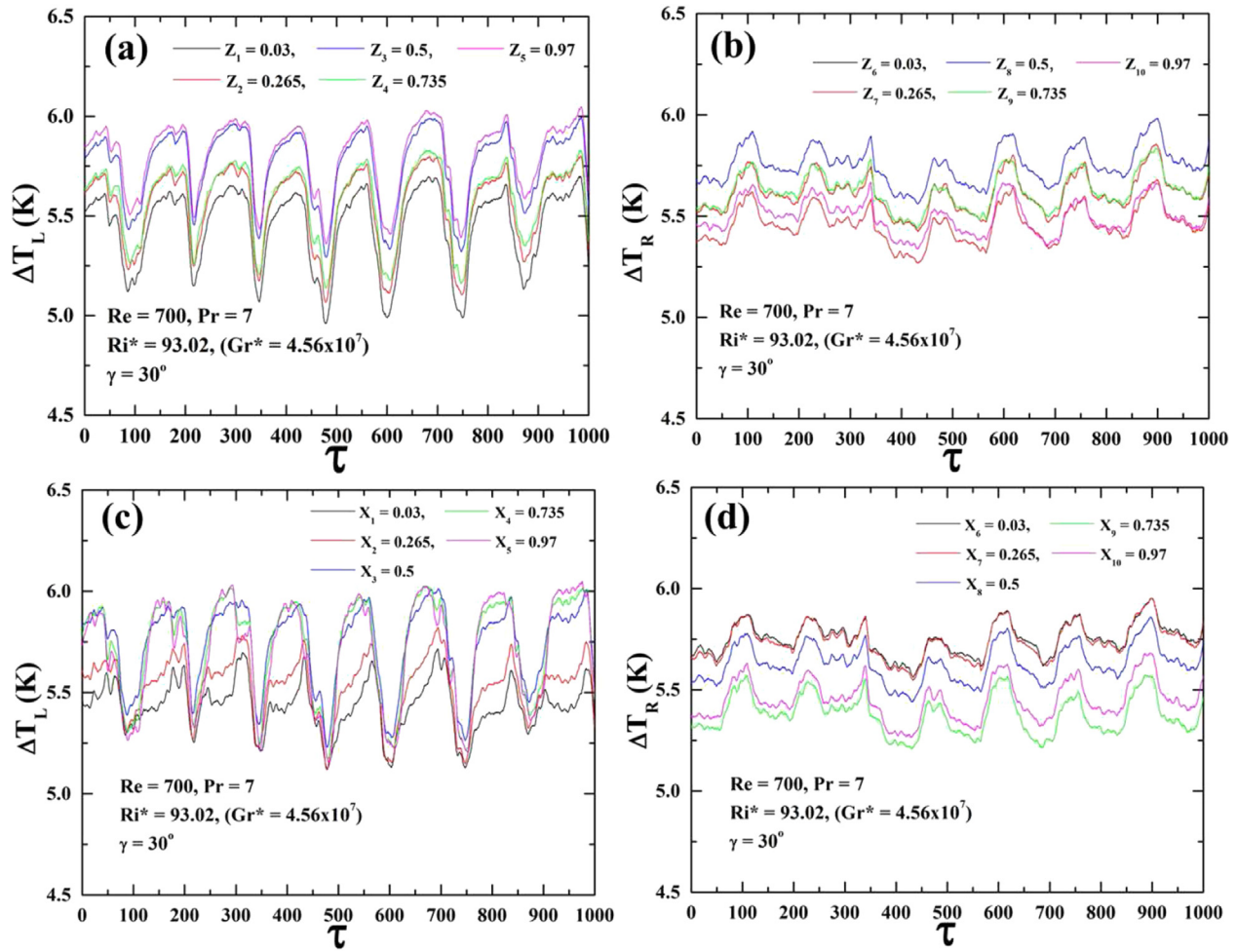


Fig. 11. Time-variations of the measured column-averaged (Z_i) and row-averaged (X_i) temperature difference between the local surface temperature and the reference temperature $\Delta T = (T_{wj} - T_0)$ along the nondimensional spanwise and axial distance for the left (ΔT_L) and right (ΔT_R) heaters at $Re = 700$, $\gamma = 30^\circ$ and $Ri^* = 93.02$.

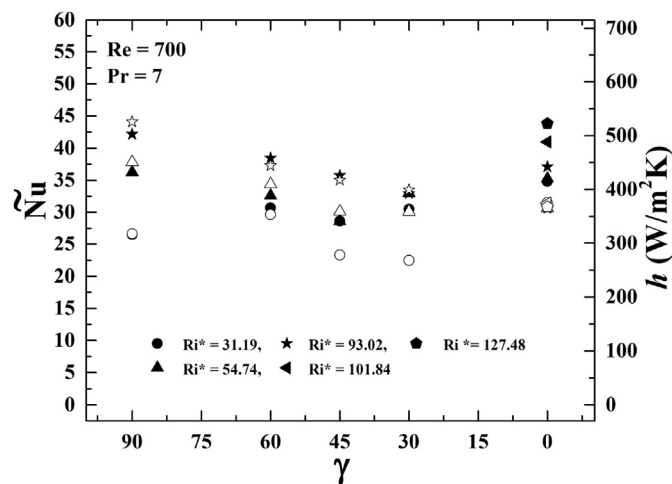


Fig. 12. Effect of the channel orientation (γ) on the thermal response for $Re = 700$ and several values of the modified Richardson number. Here, both the time-and-spaced-averaged Nusselt number and the mean convection heat transfer coefficient at each heater are plotted. The closed/open symbols correspond to the left (lower)/right (upper) heaters, respectively.

Table 1

Mean net convection heat flux per unit area transferred to the fluid as a function of the modified Richardson number at $Re = 700$.

Ri^*	0	31.19	54.74	93.02	101.84	127.48
$\bar{q}$ (W/m ²)	0	727	1276	2168	2373	2970

per unit area, convective heat transfer coefficients, overall Nusselt number distributions for each heat source, phase-space plots of the self-oscillatory system, characteristic times of temperature oscillations, and spectral distribution of the fluctuating energy have been obtained for a wide range in the parametric space. In addition, the time evolution of the temperature distributions along the discrete heat sources have also been obtained, and the results suggest several guidelines for the thermal design of electronic packages. Based on the heat transfer distributions, the following observations can be made:

1. At high opposing buoyancy, when a threshold of the buoyancy parameter is reached, transition from steady-state to time periodic or quasi-periodic flow and heat transfer takes place.
2. When a time-periodic fluctuating flow sets in, it oscillates with a fundamental frequency and the space-averaged Nusselt numbers also oscillate accordingly. Also, the amplitude of the

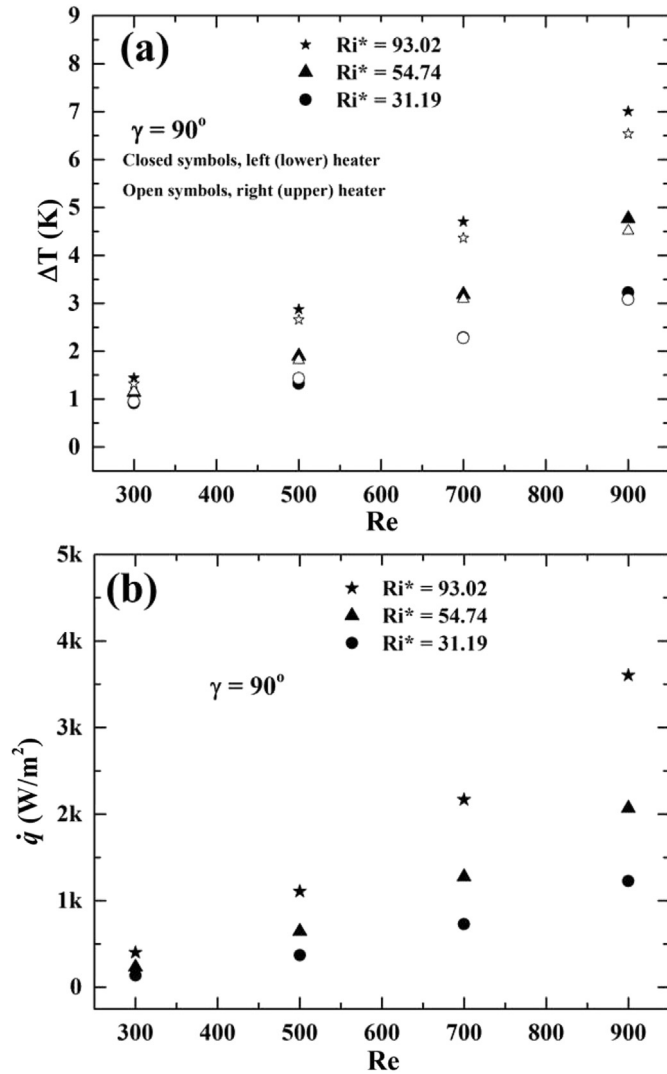


Fig. 13. (a) Measured temperature difference between the reference temperature and the mean surface temperature of each slab $\Delta T = (\bar{T}_{wj} - T_0)$ for several values of the Reynolds and modified Richardson numbers. (b) Mean net convection heat flux per unit area transferred to the fluid for several values of the Reynolds and modified Richardson numbers.

oscillations increases for increasing values of the modified Richardson number.

3. The threshold value of the modified Richardson number strongly depends on the value of the Reynolds number and duct orientation. In the limiting cases analyzed, buoyancy influences the thermal and flow responses in a two different ways. In the vertical channel configuration, buoyancy acts directly against convective flow and the vortex structure climbs towards the upper channel entrance, it changes from a two vortex structure to a complex flow structure with several interacting vortices and higher heat transfer rates are achieved because of the secondary flow. As the duct approaches the horizontal configuration, although buoyancy strength is reduced, results show that there is not a monotonic dependence between the heat transfer rates achieved for decreasing values of the inclination angle. For the horizontal configuration, because buoyancy only acts indirectly, higher threshold values of the modified Richardson number are required to induce instability.
4. For relatively high values of buoyancy strength, because of the presence of wall effects, the local Nusselt numbers present

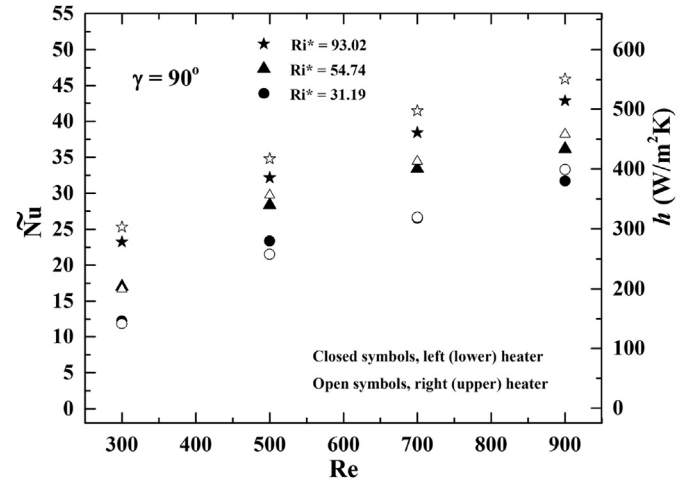


Fig. 14. Effect of the Reynolds and modified Richardson numbers on the overall heat transfer. Here, both the time-and-space averaged Nusselt number and the mean convection heat transfer coefficient at each heater are plotted as a function of the Reynolds number for $Re = 300, 500, 700$ and 900 and several values of the modified Richardson number.

strong spanwise and axial variations, elucidating important 3D behavior of the flow and heat transfer distributions.

5. In general, the values of the local Nusselt numbers are lower at the middle spanwise positions of both heaters than those registered at the other spanwise locations, indicating that the highest flow reversal of the 3D vortex structure takes place close to the channel corners and that higher surface temperatures are reached at the centerline of the discrete heat sources.

The analyzed information for opposing mixed convection in a square channel contributes to understand the transient and 3D thermal behavior of electronic chips. These results might be applicable to the analysis and design of electronic devices or equipments aimed towards heat transfer enhancement in compact heat exchangers and other cooling equipment.

Acknowledgments

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Appendix A. Relation of heater temperature and local Nusselt number

In this appendix, a relation between the surface temperature and the local heat transfer rate is to be deduced. An uniform heat flux is added to the back of an aluminum plate of thickness d_w and thermal conductivity k_w . The energy equation for the plate is

$$\frac{\partial^2 T_w}{\partial z^2} + \frac{\partial^2 T_w}{\partial y^2} = 0, \quad (\text{A.1})$$

with the boundary conditions: $\partial T_w / \partial z = 0$ at $z = 0, L$ and $\dot{q} = \dot{q}_{el}$ at $y = 0$. In the thermally thin approximation, T_w is assumed to be only function of z but with non-negligible gradients $\partial T_w / \partial y$, and then integration of Eq. (A.1) in the y direction yields

$$d_w \frac{d^2 T_w}{dz^2} + \frac{\partial T_w}{\partial y} \Big|_{y=d_w} - \frac{\partial T_w}{\partial y} \Big|_{y=0} = 0, \quad (\text{A.2})$$

or

$$d_w \frac{d^2 T_w}{dz^2} - \frac{\dot{q}}{k_w} + \frac{\dot{q}_{el}}{k_w} = 0, \quad (\text{A.3})$$

where $\dot{q}$ is the heat flux towards the fluid. Then

$$\dot{q} = \dot{q}_{el} + k_w d_w \frac{d^2 T_w}{dz^2}. \quad (\text{A.4})$$

Integration of Eq. (A.4) along the plate length, gives that

$$\frac{1}{L} \int_0^L \dot{q} dz = \dot{q}_{el}. \quad (\text{A.5})$$

However the heat flux towards the fluid is a function of z , $\dot{q}(z)$. For simplicity, it is assumed that the plate temperature can be written as a polynomial of fourth degree in the form

$$T_w(Z) = T_{wm} + \frac{c}{2} [1 - 16Z^2(1 - 2Z + Z^2)], \quad (\text{A.6})$$

where $Z = z/L$, T_{wm} is the maximum value of the temperature (assumed to be at $Z = 0.5$) and c is to be obtained as a function of the temperature difference at the plate $\Delta T_w = T_{wm} - T_w(0)$, that is $c = -2\Delta T_w$. Eq. (A.6) satisfies the boundary conditions. Therefore

$$\frac{d^2 T_w}{dz^2} = 32\Delta T_w (1 - 6Z + 6Z^2). \quad (\text{A.7})$$

Thus, Eq. (A.4) takes the form

$$\dot{q} = \dot{q}_{el} + \frac{32k_w d_w}{L^2} \Delta T_w (1 - 6Z + 6Z^2). \quad (\text{A.8})$$

The mean temperature at the plate can be obtained after integration of Eq. (A.6) along Z from 0 to 1, giving

$$\overline{T_w} = T_{wm} - \frac{7}{15} \Delta T_w. \quad (\text{A.9})$$

Therefore

$$T_w(Z) = \overline{T_w} + \Delta T_w \left[-\frac{8}{15} + 16Z^2(1 - 2Z + Z^2) \right]. \quad (\text{A.10})$$

From Eq. (A.8), the local Nusselt number is related to that computed with the input heat flux as

$$Nu = \frac{\dot{q} D_H}{k \Delta T} = \frac{\dot{q}_{el} D_H}{k \Delta T} + 32 \frac{k_w}{k} \frac{d_w}{L} \frac{D_H}{L} \frac{\Delta T_w}{\Delta T} (1 - 6Z + 6Z^2), \quad (\text{A.11})$$

where ΔT is the measured $\Delta T = T_w(Z) - T_0$. Therefore, the local Nusselt number is related to the space-averaged Nusselt number in the form

$$Nu = \overline{Nu} + \frac{\Delta T_w}{\Delta T} [\lambda G(Z) - \overline{Nu} F(Z)], \quad (\text{A.12})$$

where

$$\lambda = 32 \frac{k_w}{k} \frac{d_w}{L} \frac{D_H}{L}, \quad G(Z) = 1 - 6Z + 6Z^2, \quad F(Z) = -\frac{8}{15} + 16Z^2(1 - 2Z + Z^2). \quad (\text{A.13})$$

Using the values employed in this work, the value of λ is close to 200. Fig. A1 shows the difference $Nu - \overline{Nu}$ as a function of Z assuming the proposed temperature profile (symmetrical around $Z = 0.5$) with typical values obtained in this work. Clearly, it is shown how the local Nusselt number decreases in the middle region of the channel, where the local surface temperature is assumed to be the highest.

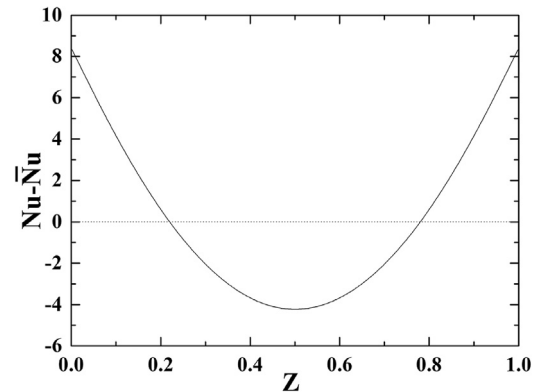


Fig. A1. Local Nusselt number variation, assuming typical values of parameters and symmetrical surface temperature profile.

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